

TextMamba3D — Project Brief

Background and Motivation: 3D brain tumor segmentation is a critical pre-step for radiotherapy planning and surgical navigation, with BraTS (4-channel MRI: T1 / T1ce / T2 / FLAIR) as the field's standard benchmark. Recent literature (e.g., TextBraTS, MICCAI 2024) has reported that feeding the paired clinical radiology report as an additional input lets SwinUNETR-class models gain about +1.5% Dice on BraTS2020, suggesting an apparently clean promise — text behaves as a "free prior" that should universally help. **However, this promise has never been rigorously validated under a unified framework:** when the backbone, the pretraining strategy, and the fusion mechanism are systematically varied, does the text gain still hold?

Research Question: Can a clinical radiology report, used as an auxiliary modality, genuinely improve 3D medical image segmentation?

Project Carrier and Contribution Definition: To answer this question under tightly controlled variables, we built TextMamba3D — a unified text-guided segmentation framework built entirely on the Mamba (SSM) architecture. We chose Mamba because its $O(n)$ complexity over 3D voxels is better suited to the 1M+ voxels of medical volumes than Transformer-style methods ($O(n^2)$), and because, to our knowledge, no prior work has unified Mamba $\times$ text guidance $\times$ 3D medical segmentation. The framework has ~60M trainable parameters, is trained on A100 / H100 GPUs, and uses BraTS2020 + TextBraTS (369 paired expert text descriptions) as the primary evaluation testbed, with BraTS2021 (1251 cases) as the pretraining source. **The core contribution is jointly defined by two hard criteria:** (1) **absolute Mean Dice strictly higher than TextBraTS SOTA (0.853), with a 1–2 percentage-point margin (i.e., ≥ 0.863)**, demonstrating that the Mamba architecture can clearly surpass SwinUNETR-class Transformer methods on 3D brain tumor segmentation; (2) **text delta significantly positive, and not lower than the +0.55% level we have already achieved on the from-scratch route**, demonstrating that text remains meaningful on top of this strong backbone. Both conditions **must hold simultaneously** — there is an intrinsic trade-off at play here: **the stronger a backbone's ability to learn image features, the more it suppresses the text contribution**, so the text delta is not required to match TextBraTS's +1.5% (that window only exists for weak backbones), but we must prove that text has not been entirely absorbed by the strong backbone.

Core Finding and the Current Bottleneck: After 10+ experimental versions, **none of our versions yet satisfies both hard criteria at the same time**. Concretely, the two main routes fail on different axes, exposing a clean zero-sum pattern:

- **From-scratch route (V5.0 / V5+V6+V7 ensemble):** text delta = +0.55%, exactly meeting the contribution floor and confirming that text is in principle effective on a Mamba backbone. But absolute Mean Dice is only **0.8479 (V5.0 single model) / 0.8511 (3-model ensemble)** — **below TextBraTS SOTA (0.853)**, and far short of the "1–2 percentage points above SOTA" target. **Criterion (1) is not met.**

- **Same-domain pretraining route (V8.0):** absolute Mean Dice jumps to **0.8753** (ET=0.8116, TC=0.8977, WT=0.9167), about 2 percentage points above SOTA — **criterion (1) is met**, which at least proves that "reaching Mamba-based absolute performance 1–2pp above SOTA on brain tumor segmentation" is in fact achievable. But the price is that the **text delta collapses precisely to 0.00%** — the text contribution is 100% absorbed by the strong backbone, and **criterion (2) is completely dead**. This is merely "a Mamba segmenter that is strong without needing text," which is by itself a counter-example to the thesis.

To rule out "maybe we just picked the wrong fusion method," we ran a systematic ablation on top of the V8.0 backbone, sequentially replacing or stacking 9 different text-utilization mechanisms (SeqCA cross-attention, MambaConcatFusion native scan fusion, 10-epoch frozen warmup, 3-stage fully frozen training, vision modality dropout, InfoNCE alignment loss, feature noise injection, decoder-side FiLM, and R-Super-style "text as supervision"). **All nine yielded text delta = 0.00%.** This points to an upstream issue: same-domain pretraining has caused the backbone to absorb, from images alone, all the information that text could possibly provide — tumor location, morphology, enhancement pattern — leaving text structurally redundant in the information-theoretic sense. This also explains why TextBraTS achieves a +1.5% text gain: its SwinUNETR is pretrained on ImageNet (cross-domain), knows nothing about brain tumors, and therefore leaves a "knowledge gap" for text to fill.

Current Experiment (V10.2): We are pretraining the backbone on BraTS2021 with self-supervised Masked Image Modeling — letting it learn only anatomical structure, with no exposure to tumor segmentation labels — to deliberately create a "knows-anatomy, does-not-know-tumors" weak backbone, so that text can fill this tumor knowledge gap during downstream fine-tuning. This is our current and final attempt to break the "strong backbone × effective text" zero-sum.

Next Steps: V10.2's success criterion strictly mirrors the two hard criteria of the core contribution — **absolute Mean Dice ≥ 0.863 (at least 1 percentage point above TextBraTS SOTA)** and **text delta $\geq +0.55\%$ (no lower than what the from-scratch route has already achieved)** — both must hold simultaneously. V8.0 has already proven that the absolute-performance axis is reachable; V5.x has already proven that the text-delta axis is reachable on a weak backbone; what V10.2 must do is **make the two coexist on a single model for the first time**. If V10.2 succeeds, the "strong backbone × effective text" zero-sum is broken, TextMamba3D becomes the *first text-guided Mamba framework strictly beating TextBraTS SOTA on both axes*, and we move into paper writing. If V10.2 fails, we will need to reframe the project's claim boundary — from "proving text is effective" to "characterizing when text becomes ineffective."

TextMamba3D 项目简介

背景与动机：3D 脑瘤分割是放疗规划与手术导航的关键前置任务，BraTS（4 通道 MRI：T1 / T1ce / T2 / FLAIR）是该方向的标准评测基准。近两年文献（如 TextBraTS, MICCAI 2024）报道：把每例 MRI 配套的临床放射报告作为额外输入，可以让 SwinUNETR 类模型在 BraTS2020 上多拿到约 +1.5% 的 Dice 增益，构成一个看似清晰的承诺——文本作为“免费的先验”应该普遍有用。但这一承诺从未在统一框架下被严格验证过：当 backbone 改变、预训练策略改变、融合机制改变时，文本增益是否仍然成立？

研究问题：临床放射报告作为辅助模态，能不能真正改善 3D 医学图像分割？

项目载体与核心贡献定义：为了在严格控制变量的前提下回答这个问题，我们搭了一个完全基于 Mamba (SSM) 架构的统一文本引导分割框架——TextMamba3D。选 Mamba 是因为它对 3D 体素具有 $O(n)$ 复杂度，比 Transformer 类方法 ($O(n^2)$) 更适合 1M+ 体素的医学体积，且目前文献中没有 Mamba × 文本引导 × 3D 医学分割的统一研究。框架约 60M 可训参数，在 A100 / H100 上训练，以 BraTS2020 + TextBraTS（369 例配对专家文本）作为评测载体，BraTS2021（1251 例）作为预训练源。项目的核心 contribution 由两个硬指标共同定义：(1) 绝对 Mean Dice 严格高于 TextBraTS SOTA (0.853)，且高出 1~2 个百分点（即 ≥ 0.863 ），证明 Mamba 架构可以在 3D 脑瘤分割上明确超过 SwinUNETR 类 Transformer 方法；(2) text delta 显著为正、且不低于我们从零训练路线已经达到的 +0.55% 水平，证明文本在这个强 backbone 上仍然有意义。两个条件必须同时满足——这背后有一个内在的 trade-off 在起作用：backbone 学习 image feature 的能力越强，对文本贡献的压制就越强，所以 text delta 不被要求匹配 TextBraTS 的 +1.5%（那是弱 backbone 才有的窗口），但必须证明文本没被强 backbone 完全吞掉。

核心发现与目前面临的问题：穿过 10+ 个实验版本之后，我们目前没有任何一个版本同时满足上述两个硬指标。具体地，两条主要路线分别在不同的轴上失败，呈现出一个干净的零和现象：

- **从零训练路线（V5.0 / V5+V6+V7 ensemble）：**text delta = +0.55%，恰好达到 contribution 的下限，证明文本对 Mamba backbone 在原理上是有效的；但绝对 Mean Dice 只有 0.8479（V5.0 单模型） / 0.8511（三模型 ensemble）——低于 TextBraTS SOTA 0.853，更够不到“高出 1~2 个百分点”的硬指标。条件 (1) 没达到。
- **同域预训练路线（V8.0）：**绝对 Mean Dice 跃升到 0.8753（ET=0.8116, TC=0.8977, WT=0.9167），高出 SOTA 约 2 个百分点，条件 (1) 达到了——这至少证明了“用 Mamba 在脑瘤分割上拿到超过 SOTA 1~2pp 的绝对性能”是可行的。但代价是 text delta 精确退化到 0.00%——文本贡献被强 backbone 100% 吞掉，条件 (2) 完全死掉。这只是“我们训了一个不需要文本就很强的 Mamba 分割器”，本身就是 thesis 的反例。

为了排除"是不是某个融合方法没选对"的可能，我们在 V8.0 backbone 上做了系统性消融——依次替换或叠加了 9 种文本利用机制（SeqCA 交叉注意力、MambaConcatFusion 原生扫描融合、10 epoch 冻结预热、3 阶段完全冻结训练、视觉模态 dropout、InfoNCE 对齐损失、特征噪声注入、解码器侧 FiLM、R-Super 风格的"文本作为监督"），**九种全部得到 text delta = 0.00%**。这指向一个上游问题：同域预训练让 backbone 已经从图像中学尽了文本本应提供的所有信息——肿瘤位置、形态、增强模式——文本因此在信息论意义上变成结构性冗余。这也解释了为什么 TextBraTS 能拿 +1.5% 的文本增益——它的 SwinUNETR 用 ImageNet（跨域）预训练，对脑瘤一无所知，给文本留下了"知识缺口"可以填补。

当前实验 (V10.2)：用自监督 Masked Image Modeling 在 BraTS2021 上预训练 backbone——只让它学解剖结构、不让他接触肿瘤分割标签——从而创造一个"懂解剖、不懂肿瘤"的弱 backbone，让文本在下游微调时去填补这个肿瘤知识缺口。这是我们目前用来打破"强 backbone × 有效文本"零和的最后一个尝试。

下一步：V10.2 的成功判定严格对应核心 contribution 的两个硬指标——**绝对 Mean Dice ≥ 0.863 （高出 TextBraTS SOTA 至少 1 个百分点）且 text delta $\geq +0.55\%$ （不低于从零训练路线已经达到的水平）**，两者必须同时满足。V8.0 已经证明绝对性能这一轴是可达的，V5.x 已经证明 text delta 这一轴在弱 backbone 下是可达的，V10.2 要做的是**让二者第一次在同一个模型上共存**。如果走通，"强 backbone × 有效文本"的零和被打破，TextMamba3D 才真正构成"first text-guided Mamba framework strictly beating TextBraTS SOTA on both axes"这样的 contribution，进入 paper 写作阶段；如果走不通，则需要重新评估项目的 claim 边界——从"证明文本有效"转为"刻画文本何时无效"。